



Thoshith R R 250701797-2025-CSE ▾

**T2**

|              |                                 |
|--------------|---------------------------------|
| Started on   | Friday, 21 August 2026, 3:21 PM |
| State        | Finished                        |
| Completed on | Friday, 21 August 2026, 3:30 PM |
| Time taken   | 9 mins 30 secs                  |
| Marks        | 1.00/1.00                       |
| Grade        | 10.00 out of 10.00 (100%)       |

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

**Note:** No need of counter increment for declarations and scanf() and counter variable printf() statement.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```
1 | #include<stdio.h>
2 | int main(){
3 |     int n;
4 |     scanf("%d",&n);
5 |     int count=0;
6 |     for(int i=1;i<=n;++i){
7 |         count++;
8 |         if(n%i==0){
9 |             count++;
10 |
11 |         }
12 |         count++;
13 |     }
14 |     count++;
15 |     printf("%d",count);
16 | }
```

|   | Input | Expected | Got |   |
|---|-------|----------|-----|---|
| ✓ | 12    | 31       | 31  | ✓ |
| ✓ | 25    | 54       | 54  | ✓ |
| ✓ | 4     | 12       | 12  | ✓ |

Passed all tests! 

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)